

Dr. André Bauer

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Curriculum Vitae

RESEARCH INTEREST

My research is in performance engineering for AI and cloud systems, making the infrastructure behind modern AI efficient, predictable, and trustworthy.

ACADEMIC CAREER

University of Chicago Booth School of Business

VISITING ASSISTANT PROFESSOR OF ECONOMETRICS AND STATISTICS

Chicago, United States

since September 2026

Illinois Institute of Technology

ASSISTANT PROFESSOR

Chicago, United States

August 2024 — August 2026

Argonne National Laboratory

ASSOCIATED RESEARCHER

Lemont, United States

January 2023 — August 2024

University of Chicago

POSTDOCTORAL SCHOLAR (PROF. IAN FOSTER)

Chicago, United States

November 2022 — August 2024

University of Würzburg

RESEARCH GROUP LEADER (PROF. SAMUEL KOUNEV)

Würzburg, Germany

January 2021 — October 2022

University of Würzburg

RESEARCH ASSISTANT

Würzburg, Germany

October 2016 — December 2020

SCHOOL & ACADEMIC EDUCATION

University of Würzburg

HABILITATION IN COMPUTER SCIENCE

Würzburg, Germany

July 2022 — October 2026

- **Thesis** “Enabling Efficient Scientific Computing: Intelligent Resource Management in the Age of Cloud and AI”
- **Mentors** Prof. Dr. Samuel Kounev, Prof. Dr. Ingo Scholtes, Prof. Dr. Rüdiger Pryss
- **Status** Thesis submitted December 2025; final faculty decision scheduled for October 22, 2026

University of Würzburg

DR. RER. NAT. (COMPUTER SCIENCE)

Würzburg, Germany

October 2016 — December 2020

- **Dissertation** “Automated Hybrid Time Series Forecasting: Design, Benchmarking, and Use Cases”
- **Advisor** Prof. Dr. Samuel Kounev
- **Grade** Summa cum laude (1.0) — Defense December 22, 2020

University of Würzburg

MASTER OF SCIENCE IN COMPUTER SCIENCE

Würzburg, Germany

October 2014 — September 2016

Blekinge Institute of Technology

SEMESTER ABROAD (ERASMUS PROGRAM)

Karlskrona, Sweden

August 2015 — January 2016

University of Würzburg

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Würzburg, Germany

October 2012 — September 2014

AWARDS & SCHOLARSHIPS

Best Industry Paper Award, ACM/SPEC International Conference on Performance Engineering (ICPE)

Florence, Italy

For the paper “Evaluating Kubernetes Performance for GenAI Inference: From Automatic Speech Recognition to LLM Summarization”.

May 2026

Best Student Paper Award, IEEE International Conference on e-Science (eScience)

Chicago, United States

For the paper “Efficient Privacy-Preserving Recommendation on Sparse Data using Fully Homomorphic Encryption” (co-author).

September 2025

Best Teacher Award, Illinois Institute of Technology

Chicago, United States

The department of computer science honors the best teacher every year.

April 2025

GAIN Participation Grant , German Academic International Network The GAIN conference connects internationally mobile German scientists with representatives from the scientific community, politics, and industry.	Bonn, Germany May 2023
Runner-up for Best Data Challenge Paper , ACM/SPEC International Conference on Performance Engineering (ICPE) For the paper “Efficient Data Processing: Assessing the Performance of Different Programming Languages”.	Coimbra, Portugal April 2023
2023 SPECTacular Award , Standard Performance Evaluation Corporation (SPEC) The SPECTacular Award is given to individuals who have significantly exceeded expectations in their contributions to SPEC.	Gainesville, United States February 2023
Leadership Academy Fellowship , German Scholars Organization The Leadership Academy Fellowship is awarded to 25 outstanding researchers from all disciplines.	Berlin, Germany January 2023
Best Industry Paper Award , ACM/SPEC International Conference on Performance Engineering (ICPE) For the paper “Why Is It Not Solved Yet? Challenges for Production-Ready Autoscaling”.	Virtual conference April 2022
SPEC Kaivalya Dixit Distinguished Dissertation Award , SPEC The SPEC Kaivalya Dixit Distinguished Dissertation Award is an annual award recognizing one outstanding doctoral dissertation within the scope of the SPEC Research Group in terms of scientific originality, scientific significance, practical relevance, impact, and presentation.	Gainesville, United States April 2022
Distinguished Dissertation Award , Faculty of Mathematics and Computer Science The Distinguished Dissertation Award is an annual award for outstanding dissertations from the Faculty of Mathematics and Computer Science.	Würzburg, Germany July 2021
Dissertation Award , Unterfränkische Gedenkjahrstiftung für Wissenschaft The Dissertation Award is an annual award for outstanding dissertations from all disciplines.	Würzburg, Germany May 2021
Summa cum laude , University of Würzburg Grading of the doctoral thesis ‘Automated Hybrid Time Series Forecasting: Design, Benchmarking, and Use Cases’.	Würzburg, Germany December 2020
Winner of the FameLab State of Bavaria final , British Council In this science communication competition, researchers must present their work in three minutes in an understandable and humorous way.	Regensburg, Germany March 2020
Sokrates/Erasmus Grant , University of Würzburg Mobility grant for a semester abroad in Sweden.	Würzburg, Germany August 2015 — January 2016
Deutschlandstipendium , University of Würzburg The Deutschlandstipendium supports talented and high-achieving students, taking into account not only first-class grades but also social commitment and special personal achievements.	Würzburg, Germany October 2014 — September 2015
Heinrich-Thein Award , Heinrich-Thein-Stiftung Award for talented students pursuing the “second-chance education”.	Haßfurt, Germany November 2009

FUNDING

NSF CRII Grant (\$150,000) , National Science Foundation PI for the project “Co-Simulation-Based Optimization of Container Orchestration Platforms”.	August 2025
Google Research Grant (\$5,000) , Google Acquisition of Google Cloud credits for the project “Assesing Microservice Applications Performance for Benchmarking”.	May 2025
Google Research Grant (\$1,000) , Google Acquisition of Google Cloud credits for the project “Performance Analysis and Prediction of Microservices in Cloud Environments”.	March 2023
DFG Research Grant (\$347,235) , German Research Foundation (DFG) Co-PI, with Prof. Samuel Kounnev, for the project “MINERVA: Performance Prediction of Microservice Applications with Black-box Container Orchestration”.	January 2023

Industrial project, DATEV eG Assistance for the extension of the project “Self-Configuring Autoscaler” (\$30,068), with a project duration of 3 months.	May 2022
Industrial project, DATEV eG Assistance with the application of the project “Self-Configuring Autoscaler” (\$71,200), with a project duration of 6 months.	July 2021
Normalverfahren Grant 2021 (\$3,234), Universitätsbund Würzburg Acquisition of the Normalverfahren Grant for the project “Prediction of Winter Mortality of Bee Colonies Using Machine Learning”.	April 2021

TEACHING ACTIVITIES

Analysis of Financial Time Series (BUSN 41921), University of Chicago Booth School of Business Lecturer of this MBA course (English), Spring Quarter 2027.	Chicago, United States March 2027 — June 2027
Advanced Business Statistics (BUSN 41001), University of Chicago Booth School of Business Lecturer of this MBA course (English), Winter Quarter 2027.	Chicago, United States January 2027 — March 2027
Time Series Analysis and Forecasting, Illinois Institute of Technology Lecturer of this self-created master’s course (English).	Chicago, United States January 2026 — May 2026
Operating Systems, Illinois Institute of Technology Lecturer of this bachelor’s and master’s course (English).	Chicago, United States August 2025 — December 2025
Time Series Analysis and Forecasting, Illinois Institute of Technology Lecturer of this self-created master’s course (English).	Chicago, United States January 2025 — May 2025
Distributed Systems, University of Chicago Co-Lecturer with Prof. Kyle Chard of this master’s course (English).	Chicago, United States September 2023 — December 2023
Time Series Analysis and Forecasting, University of Würzburg Lecturer of this self-created master’s course (English).	Würzburg, Germany April 2022 — September 2022
Advanced Programming, University of Würzburg Co-Lecturer with Prof. Samuel Kounev of this master’s course (English).	Würzburg, Germany October 2021 — March 2022
Computer Science: Getting Started, University of Würzburg Co-Lecturer with Prof. Samuel Kounev of this bachelor’s course (German).	Würzburg, Germany April 2021 — September 2021
Operating Systems, University of Würzburg Co-Lecturer with Dr. Nikolas Herbst of this bachelor’s/master’s course (German).	Würzburg, Germany October 2020 — March 2021
Software-Engineering Seminar, University of Würzburg Instructor of this bachelor’s/master’s course (German).	Würzburg, Germany October 2016 — March 2021
Operating Systems (Exercise), University of Würzburg Coordinator & instructor of this bachelor’s/master’s course (German).	Würzburg, Germany October 2016 — March 2021
Practical Course in Software, University of Würzburg Coordinator & instructor of this bachelor’s course (German).	Würzburg, Germany October 2016 — September 2020

INVITED TALKS

An Open Function-as-a-Service Dataset From the Edge to the Cloud PARSL FEST	Chicago, US September 2024
Academic Success: Navigating your Journey UNIVERSITY OF CHICAGO	Chicago, US February 2024
Smart Infrastructure Management: Navigating Efficiency in Scientific Computing ILLINOIS INSTITUTE OF TECHNOLOGY	Chicago, US November 2023
Why Auto-Scaling Still Matters DEPAUL UNIVERSITY	Chicago, US October 2023

Towards an Intelligent Infrastructure for Data Science Clouds

CHALMERS UNIVERSITY OF TECHNOLOGY

Edmonton, Canada

May 2023

Autonomous Data Analytic Clouds

CHALMERS UNIVERSITY OF TECHNOLOGY

Gothenburg, Sweden

February 2023

A Short Overview of Auto-Scaling & Why It Is Not Yet Solved

MONGODB

Virtual meeting

April 2022

Measuring Energy-Efficiency in Digital Systems

HASSO-PLATTNER-INSTITUT

Potsdam, Germany

March 2022

ACADEMIC SERVICE

Organization Committee

- Workshop on Hot Topics in Cloud Computing Performance 2025
- International Conference on Performance Engineering (Data Challenge) 2025
- International Conference on e-Science (Posters) 2024
- International Conference on Performance Engineering (Posters and Demos) 2024
- International Workshop on Performance-Data Analytics and Data-Management 2023, 2022
- International Conference on Utility and Cloud Computing 2026, 2025, 2024, 2023, 2022
- International Conference on Time Series and Forecasting 2026, 2025, 2024, 2023, 2022
- International Conference on Data Science, Technology and Applications 2026, 2025, 2024, 2023
- Int'l Conference on Autonomic Computing and Self-Organizing Systems 2026, 2025, 2024
- International Conference on e-Science 2026, 2025, 2024
- International Conference on Performance Engineering 2027, 2026, 2025, 2024
- International Workshop on Intelligent and Adaptive Edge-Cloud Operations and Services 2026, 2025
- International Conference for High Performance Computing, Networking, Storage and Analysis 2026
- International Conference on Software Engineering for Adaptive and Self-Managing Systems 2027, 2026
- Workshop on Hot Topics in Cloud Computing Performance 2026
- Symposium on Software Performance 2025, 2024, 2023, 2022
- International Symposium on Cluster, Cloud and Internet Computing (Artifact Evaluation) 2025, 2024
- International Conference on Service-Oriented System Engineering 2025
- International Conference on Cloud Computing and Services Science 2024
- International Symposium on Cluster, Cloud and Internet Computing 2023
- Int'l Conference on Autonomic Computing and Self-Organizing Systems (Posters and Demos) 2023, 2022
- International Conference on Performance Engineering (Artifact Evaluation) 2020
- Workshop on Evaluations and Measurements in Self-Aware Computing Systems 2019

Associate Editor

- Frontiers in Digital Health, Health Informatics since March 2026

Editorial Board Member

- ICCK Journal of Systems Scalability since October 2025
- MDPI Data since January 2025

Financial Chair

- International Conference on e-Science 2025

Publicity Chair

- International Conference on Performance Engineering 2020
- Workshop on Evaluations and Measurements in Self-Aware Computing Systems 2019

Standard Performance Evaluation Corporation

- Founder and chair of the SPEC RG Predictive Data Analytics Working Group since June 2021
- Elected editor of the annual SPEC RG newsletter April 2017 — May 2023
- Elected release manager of the SPEC RG Cloud Working Group October 2018 — May 2021

Journal Reviews

- Journal of Systems & Software 2026, 2022, 2018
- Journal of Parallel and Distributed Computing 2026, 2021, 2019
- Journal of Grid Computing 2026, 2023
- Journal of Systems Scalability 2026
- Performance Evaluation 2026
- Journal of Supercomputing 2025
- Information Processing and Management 2025
- Frontiers Digital Health 2025
- Empirical Software Engineering 2024, 2023, 2022
- SoftwareX 2024

• Transactions on Knowledge and Data Engineering	2023
• Science of Computer Programming	2023
• Transactions on Autonomous and Adaptive Systems	2023
• Transactions on Cloud Computing	2021
• Transactions on Services Computing	2020
• Journal of Future Generation Computer Systems	2019
• Transactions on Network and Service Management	2018

SCIENCE COMMUNICATION COMPETITIONS

1st South Side Science Festival Science Slam UNIVERSITY OF CHICAGO	<i>Chicago, USA</i> <i>September 2023</i>
6th GAIN Science Slam GERMAN ACADEMIC INTERNATIONAL NETWORK	<i>Boston, USA</i> <i>August 2023</i>
21st Stuttgart Science Slam LUUPS	<i>Stuttgart, Germany</i> <i>October 2022</i>
FameLab State of Bavaria final BRITISH COUNCIL	<i>Regensburg, Germany</i> <i>March 2021</i>
6th Alumni-Science Slam UNIVERSITY OF WÜRZBURG	<i>Würzburg, Germany</i> <i>November 2020</i>
FameLab German final BRITISH COUNCIL	<i>Bielefeld, Germany</i> <i>May 2020</i>
FameLab State of Bavaria final BRITISH COUNCIL	<i>Regensburg, Germany</i> <i>March 2020</i>
14th Munich Science Slam LUUPS	<i>Munich, Germany</i> <i>February 2020</i>
33rd Dortmund Science Slam LUUPS	<i>Dortmund, Germany</i> <i>November 2019</i>

CERTIFICATES

Certificate of Leadership – Smart Skills National Postdoctoral Association	<i>January 2024</i>
Certificate Leadership University of Würzburg	<i>October 2022</i>
Certificate Academic Teaching – State of Bavaria University of Würzburg	<i>November 2020</i>
Certificate Academic Teaching – University of Würzburg University of Würzburg	<i>December 2019</i>

VOLUNTARY ACTIVITIES

Mentoring for high school students ILLINOIS SCIENCE & TECHNOLOGY COALITION	<i>Chicago, United States</i> <i>since August 2024</i>
Pen Pal Program OFFICE OF DIVERSITY AND INCLUSION UNIVERSITY OF CHICAGO	<i>Chicago, United States</i> <i>February 2023 — August 2024</i>
Mentoring for students ALUMNI NETWORK UNIVERSITY OF WÜRZBURG	<i>Würzburg, Germany</i> <i>November 2019 — October 2022</i>
Supervision of scouts SCHWARZER HAUFEN WÜRZBURG (BDP E.V.)	<i>Würzburg, Germany</i> <i>July 2019 — February 2020</i>
Volunteer tutoring NACHBARSCHAFTSHILFE GROMBÜHL	<i>Würzburg, Germany</i> <i>September 2013 — September 2014</i>
Student council FRIEDRICH-FISCHER SCHULE	<i>Schweinfurt, Germany</i> <i>September 2010 — August 2011</i>
Volunteer fire fighter FFW HORHAUSEN E.V.	<i>Horhausen, Germany</i> <i>April 2003 — August 2009</i>

LANGUAGES

German Mother tongue
English Fluent